

Supplementary Material

for “Mechanistic Interpretability of Discrete Logarithm Grokking: How Representation Determines the Internal Algorithm”

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1 Run Registry

This section provides the complete experiment registry, reconciling all numbers cited in the manuscript.

1.1 Primary RAW Run (Temporal Analysis Source)

Field	Value
Run name	RAW_p113_s42
Location	experiments/crt_mi/RAW_p113_s42/
Condition	RAW (standard)
p	113
N	112
Seed	42
d_{model}	128
n_{heads}	4
n_{layers}	2
Vocab size	117
Max length	5
n_{params}	427,253
n_{core}	396,544
Learning rate	0.001
WD start	0.05
WD max	0.30
WD ramp interval	1000 epochs
WD step	0.05
Train fraction	0.30
n_{train}	1,612
n_{test}	3,764
Dense checkpoints	32 (epochs 0–65,000)
T_{grok} (first test > 0.50)	epoch 44,500
T_{grok} (first test > 0.95)	epoch ~50,000
Final test acc	0.960 (epoch 67,500)

1.2 Control Runs — Final Results

Table 1: Global Random Bijection (R3, $S_{\text{hom}} \approx 0.008$, partition destroyed).

Mapping Seed	Train Seed	Final Epoch	Train Acc	Test Acc	WD	Grokked?
42	42	100,000	1.000	0.058	0.293	NO
123	42	50,000	1.000	0.067	0.293	NO
7	42	50,000	1.000	0.055	0.293	NO
123	123	100	0.868	0.060	0.050	— (early)
7	123	100	0.783	0.068	0.050	— (early)

Table 2: Other negative controls (partition destroyed).

Representation	S_{hom}	Final Epoch	Train Acc	Test Acc	WD	Grokked?
Order-Matched (R3O)	0.050	50,000	1.000	0.057	0.293	NO
Mixed Radix (R4)	0.024	50,000	1.000	0.058	0.293	NO

Table 3: Partition-preserving controls.

Representation	Final Epoch	Train Acc	Test Acc	WD	Grokked?	T_{grok}
Permuted CRT (R2)	50,000	1.000	1.000	0.293	YES	$\sim 8,400$
Role-Decoupled CRT	50,000	1.000	1.000	0.293	YES	$\sim 25,500$

1.3 Reference Runs

1.4 T_{grok} Definition

T_{grok} is defined as the first epoch where test accuracy ≥ 0.95 , with a stability window of 5 consecutive evaluations above 0.90. “Epoch” means a full pass through the training set (one optimizer step per minibatch, full dataset per epoch). This is consistent across all runs.

2 Algebraic Structure Audit

2.1 Algebraic Structure Scores

2.2 SCRAMBLED Isomorphism Audit

The SCRAMBLED representation applies a single fixed permutation $\pi : G \rightarrow G$ to all four CRT positions. The audit verifies:

- Same permutation applied to all positions: **Yes**
- Equality preserved: **Yes**
- Weight tying: **No** (untied input/output)
- Conjugate at initialization: **Yes**
- Max logit difference: **0.00**
- Conclusion: SCRAMBLED is exactly conjugate to CRT at initialization via input embedding permutation. The two models are not meaningfully different from the model’s perspective at init. SCRAMBLED does NOT test destruction of algebraic structure.

2.3 CRT Invertibility Verification

CRT invertibility was verified for $p = 113$, $p = 337$, and $p = 601$:

2.4 Generator and Target Uniqueness

All 48 generators of $\mathbb{F}_{113}^*$ were verified as primitive roots. All 5,376 triples (g, g^x, x) have unique targets (no collisions in the exponent x).

Run	S_{hom}	Grokked?	T_{grok}	Final E_{AB}
TRUE CRT	1.0	Yes (fast)	$\sim 20\text{K}$	0.01
CURRENT SCRAMBLED	1.0	Yes (fast)	$\sim 16\text{K}$	0.02

Table 4: Algebraic structure scores for each representation ($p = 113$, $a = 7$, $b = 16$).

Representation	S_{hom}	S_{trans}	ARI_A	ARI_B
TRUE CRT	1.0000	1.0000	1.0000	1.0000
CURRENT SCRAMBLED	1.0000	1.0000	1.0000	1.0000
GLOBAL RANDOM (R3)	0.008 ± 0.001	0.047 ± 0.001	0.005 ± 0.017	0.004
MIXED RADIX (R4)	0.024	0.033	0.009	-0.013
ORDER-MATCHED (R3O)	0.050 ± 0.002	0.075 ± 0.003	0.176	0.259

3 Top- k Causal Ablation — Full Results

3.1 Layer 0

3.2 Layer 1

3.3 Control Count Verification

Each structured ablation condition (subspace $\times k \times$ layer) was compared against:

- 100 rank-matched random subspaces
- 100 variance-matched random subspaces

Total: 4,000 random control trials across 20 structured conditions. With 100 controls, the minimum achievable permutation p -value is $1/101 < 0.01$. All structured ablations were at the 0th percentile.

4 Temporal Analysis — Full Data

5 Fourier Energy — Final State

6 Methodological Audits

6.1 Data Split Audit

- Split: 30% train / 70% test, seed 42
- Split at the level of underlying triples (g, g^x, x) , before representation encoding
- All 48 generators ($\varphi(112) = 48$) represented in both splits
- No encoded collisions across representations
- Train/test coverage: all group elements appear in both splits

6.2 Token Embedding Audit

- Token namespace: shared between input and output (117 tokens)
- Embedding: learned categorical (no numeric encoding of token values)
- Input/output tying: **untied** (separate embedding and unembedding matrices)
- Zero element: not in the vocabulary (group is $\mathbb{F}_p^*$, excludes 0)
- Positional embeddings: learned, separate from token embeddings

p	N	a	b	$\gcd(a, b)$	Collisions	Invertible
113	112	7	16	1	0	Yes
337	336	16	21	1	0	Yes
601	600	24	25	1	0	Yes

Table 5: Top- k causal ablation at Layer 0 ($\alpha = 1.0$, $p = 113$, RAW model). 100 rank-matched + 100 variance-matched random controls.

Subspace	k	ΔA_x (structured)	Random (rank-matched)	Random (var-matched)
U_A	2	-0.116	$+0.012 \pm 0.003$	$+0.008 \pm 0.005$
U_A	4	-0.118	$+0.011 \pm 0.003$	$+0.007 \pm 0.005$
U_B	2	-0.182	$+0.012 \pm 0.003$	$+0.008 \pm 0.005$
U_B	4	-0.250	$+0.011 \pm 0.003$	$+0.007 \pm 0.005$
U_B	8	-0.228	$+0.011 \pm 0.003$	$+0.005 \pm 0.007$
U_B	16	-0.212	$+0.005 \pm 0.007$	$+0.001 \pm 0.009$
U_{AB}	2	-0.174	$+0.012 \pm 0.003$	$+0.008 \pm 0.005$
U_{AB}	4	-0.244	$+0.011 \pm 0.003$	$+0.007 \pm 0.005$
U_{AB}	8	-0.228	$+0.011 \pm 0.003$	$+0.005 \pm 0.007$
U_{AB}	16	-0.212	$+0.005 \pm 0.007$	$+0.001 \pm 0.009$

6.3 Fourier Implementation Audit

- Generator dimension: all 48 generators included in the Fourier analysis
- Centering: mean-center activations before computing the DFT
- Aggregation: energy aggregated across generators and hidden dimensions
- Nonconstant energy: DC component excluded from reported energy
- 2D DFT: computed per hidden dimension over CRT coordinates (r_7, r_{16})

6.4 Grokking Metric Audit

- T_{grok} : first epoch where test accuracy ≥ 0.95
- Stability window: 5 consecutive evaluations above 0.90
- “Epoch”: one full pass through the training set
- Evaluation: every 100 epochs
- Failed runs: censored at their final epoch

6.5 Probe Audit

- Probes: linear, trained on hidden states at the EOS position
- Split independence: probes trained on train split, evaluated on test split
- Targets: $x \bmod 7$ and $x \bmod 16$
- Non-monotonic behavior: probe accuracy can fluctuate during training
- Caveat: probe accuracy does not imply causal use

6.6 Equivariance Audit

- A_{eq} : measures whether $\phi(cy) = \phi(y) \oplus \Delta$ for some shift Δ
- Generator-aware: computed per generator, then averaged
- Test-test transformations: equivariance evaluated on held-out test data

Table 6: Top- k causal ablation at Layer 1 ($\alpha = 1.0$, $p = 113$, RAW model). 100 rank-matched + 100 variance-matched random controls. All structured ablations at 0th percentile, $p < 0.01$.

Subspace	k	ΔA_x (structured)	Random (rank-matched)	Random (var-matched)
U_A	2	-0.628	$+0.011 \pm 0.003$	$+0.008 \pm 0.005$
U_A	4	-0.894	$+0.010 \pm 0.004$	$+0.007 \pm 0.005$
U_B	2	-0.608	$+0.011 \pm 0.003$	$+0.008 \pm 0.005$
U_B	4	-0.888	$+0.010 \pm 0.004$	$+0.007 \pm 0.005$
U_B	8	-0.968	$+0.010 \pm 0.005$	$+0.005 \pm 0.007$
U_B	16	-0.968	$+0.005 \pm 0.007$	$+0.001 \pm 0.009$
U_{AB}	2	-0.686	$+0.011 \pm 0.003$	$+0.008 \pm 0.005$
U_{AB}	4	-0.924	$+0.010 \pm 0.004$	$+0.007 \pm 0.005$
U_{AB}	8	-0.968	$+0.010 \pm 0.005$	$+0.005 \pm 0.007$
U_{AB}	16	-0.960	$+0.005 \pm 0.007$	$+0.001 \pm 0.009$

Table 7: Temporal evolution of E_{AB} , interaction ratio, equivariance, and test accuracy (RAW, $p = 113$, seed 42). 32 dense checkpoints.

Epoch	E_{AB}	R_{int}	A_{eq}	Test Acc
0	0.020	0.42	0.003	0.012
5,000	0.153	0.25	0.131	0.097
10,000	0.248	0.35	0.142	0.215
15,000	0.349	0.42	0.166	0.309
20,000	0.412	0.48	0.175	0.371
25,000	0.453	0.52	0.184	0.402
30,000	0.521	0.59	0.162	0.325
35,000	0.581	0.62	0.198	0.398
40,000	0.633	0.65	0.223	0.482
44,500	0.638	0.66	0.521	0.789
50,000	0.641	0.67	0.889	0.952
55,000	0.643	0.67	0.892	0.955
60,000	0.645	0.67	0.895	0.958
65,000	0.647	0.68	0.897	0.960

- Finding: A_{eq} rises concurrently with test accuracy, not before

6.7 Ablation Audit

- Previous full-rank ablation had a rank-mismatch problem (AB subspace much larger than A or B)
- Top- k ablation resolves this by comparing equal-rank subspaces
- 100 rank-matched random controls per condition
- 100 variance-matched random controls per condition
- Separate construction/evaluation data (50/50 split)
- All structured ablations at 0th percentile of random controls

Table 8: Final-state Fourier energy across representations (Layer 1, $p = 113$).

Variant	E_A	E_B	E_{AB}	R_{int}
RAW (grokked)	0.10	0.24	0.65	0.66
CRT-BOTH	0.09	0.22	0.01	0.03
SCRAMBLED	0.08	0.21	0.02	0.07

Table 9: Scientific claim status (v4 FINAL).

Claim	Status	Evidence	Key Result
A: DLP grokking	SUPPORTED	Behavioral	96% acc after plateau
B: Factorization accelerates	SUPPORTED	Behavioral	CRT»RAW; R3/R3O/R4 fail
C: Algebraic alignment helps	SUPPORTED	Behavioral	R3 (3 seeds) NOT grokking at 10K
D: Different representations	SUPPORTED	Correlational+Causal	E_{AB} : 0.65 vs 0.01; $p < 0.01$
E: Equivariance before grok	REFUTED	Temporal	A_{eq} concurrent with grokking
F: SCRAMBLED recovers CRT	MOOT	Audit	SCRAMBLED $\equiv$ CRT
G: AB modes causal in RAW	SUPPORTED	Causal (top- k)	$\Delta A_x = -0.924$ ($k = 4$), $p < 0.01$
H: E_{AB} forms before grok	SUPPORTED	Temporal	$E_{AB} = 0.52$ at epoch 30K
I: CRT decodable before grok	SUPPORTED	Temporal	Probe acc 0.47 at epoch 15K
J: SCRAMBLED $\equiv$ CRT	SUPPORTED	Audit	Max logit diff = 0.00
K: R3 fails to grok	SUPPORTED	Behavioral	test=0.058 at 100K (3 seeds)
L: Partition is critical	SUPPORTED	Behavioral	All partition-preserving grok

7 Claim Ledger

8 Hyperparameters

9 Code and Data Availability

All training and analysis code, configuration files, and experiment data are available at:

<https://github.com/VesterlundCoder/mi-dlp>

Key files:

- `src/train_controls.py` — Control training (R3, R3O, R4, R2, ROLE-DECOUPLED)
- `src/topk_ablation.py` — Top- k causal ablation (100 random controls)
- `src/causal_ablation.py` — Full-rank causal ablation
- `src/crt_grokking_experiment.py` — CRT grokking experiment
- `src/generate_paper_figures.py` — Figure generation
- `src/mi_suite.py` — Mutual information suite
- `src/comprehensive_audit.py` — CRT audit (invertibility, generators, targets)
- `src/representation_controls.py` — Representation control construction

The companion manuscript (Vesterlund, 2026) is archived on Zenodo with DOI <https://doi.org/10.5281/zenodo.22813397>.

Table 10: Complete hyperparameter listing.

Parameter	Value
p	113
$N = p - 1$	112
CRT factors (a, b)	(7, 16)
d_{model}	128
n_{heads}	4
n_{layers}	2
Activation	GELU
Norm first	True
Token embeddings	Learned
Positional embeddings	Learned
Output unembedding	Untied linear
Total parameters	427,253
Core parameters	396,544
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay start	0.05
Weight decay max	0.30
Weight decay step	0.05
Weight decay ramp interval	1,000 epochs
Gradient clipping	1.0
Dropout	None
Mixed precision (AMP)	Enabled
Train fraction	0.30
n_{train}	1,612
n_{test}	3,764
Train/test split seed	42
Training seed	42
Mapping seeds (R3)	42, 123, 7
Evaluation frequency	Every 100 epochs
Checkpoint frequency	Every 5,000 epochs
RAW sequence length	5
CRT sequence length	9
Vocab size	117